CLINICAL PHARMACY WORKSPACE

PharmaBridge

System Architecture, Functional Specifications, and User Operational Manual

An AI-assisted, clinically guarded prescription fulfilment workspace engineered for Indian community and retail pharmacies. Integrates formulation equivalence matching, transparent whole-pack economics, mandatory pharmacist clinical gates, and FEFO inventory deduction.

Document Version:	1.0 (Production-Grade Release)
System Release:	PharmaBridge v1.0.0-MVP
Backend Architecture:	FastAPI (Python 3.12) + PostgreSQL 17 (asyncpg)
Frontend Architecture:	React 19 + TypeScript + Vite + Tailwind CSS
Verification Suite:	21 Pytest Tests + 12 Playwright E2E Tests (100% Passed)
Target Jurisdiction:	India (Pharmacy Act 1948, Drugs & Cosmetics Rules)

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1. Executive Summary & Primary Mission

1.1 What is the Main Goal of PharmaBridge?

In Indian retail and community pharmacies, dispensing prescription medication is a high-stakes, time-sensitive process fraught with clinical, financial, and operational challenges. Thousands of pharmaceutical manufacturers market identical or near-identical generic formulations under brand names varying significantly in price, packaging, and availability.

The primary mission of **PharmaBridge** is to provide a unified, highly guarded digital workstation that empowers pharmacists to safely, efficiently, and transparently fulfil prescriptions. PharmaBridge bridges the gap between raw doctor prescriptions (handwritten, scanned, or digital) and actual pharmacy inventory, ensuring:

- **Zero Clinical Compromise:** Every suggested alternative brand or generic substitution must be strictly bio-equivalent in active ingredient, canonical salt identity, strength, dosage form, and release profile (Immediate Release vs. Extended Release).
- **Pack Economics Transparency:** Indian drug regulations and blister packaging often require dispensing whole packs. PharmaBridge computes transparent whole-pack economics, exposing exact patient cost savings without misleading fractional-pill pricing.
- **Mandatory Pharmacist Clinical Authority:** Only licensed pharmacists can approve substitutions or finalize orders. Assistants and technicians are gated from clinical decisions.
- **FEFO Inventory Integrity:** Inventory deduction is strictly First-Expired, First-Out (FEFO), with transactional database locks preventing stock race conditions and waste.
- **Resilient Offline-First Architecture:** Full operation is guaranteed using synthetic demo fixtures and local database persistence even if cloud OCR or external APIs fail.

> **Core Mission Statement:** To deliver an intelligent, mathematically precise, and clinically safe pharmacy dispensing workflow that lowers medicine costs for Indian patients while protecting pharmacy operators from dispensing errors and regulatory non-compliance.

2. The Problem Space & PharmaBridge Solution

To understand why PharmaBridge was built, consider the traditional manual workflow in typical Indian pharmacies versus the PharmaBridge digital pipeline:

Challenge Area	Traditional Pharmacy Practice	The PharmaBridge Solution
Prescription Legibility & Intake	Manual reading of doctor handwriting; frequent confusion over similar brand names (e.g., Metformin IR vs. Metformin ER).	Multi-format OCR intake (images/PDFs) coupled with a required manual line verification step. Ambiguities are highlighted in amber and must be resolved before proceeding.
Generic Substitution Risk	Ad-hoc substitution driven by wholesaler profit margins rather than clinical equivalence; risk of salt mismatches or partial combination errors.	Deterministic formulation matching engine. Automatically filters candidates by canonical salt, release profile, and route. Formally rejects partial combination subsets and narrow therapeutic index drugs.
Pricing & Pack Economics	Calculating pill prices on manual paper scraps; patients surprised by whole-pack costs or buying unnecessary excess units.	Exact Decimal paise arithmetic (zero floating-point drift). Transparently breaks down requested units, packs needed, excess units, and delivers demonstrable cost comparisons (e.g. 33.33% real savings).
Staff Responsibility & Gating	Untrained shop assistants often dispense medications without pharmacist supervision; no digital audit trail.	Role-based permissions (Pharmacist vs. Assistant vs. Admin). Only authenticated Pharmacists can sign off on clinical substitutions and trigger final stock deduction.
Stock Waste & Expiry	Disorganized shelves lead to newer batches being dispensed while older stock expires in the back (LIFO or random dispensing).	Automated FEFO (First-Expired, First-Out) batch allocation. Batches nearing expiry are prioritized; quarantined or expired stock is strictly blocked.

3. Core System Features & Functional Modules

PharmaBridge is structured into tightly integrated functional modules:

1. Intake & Document Processing

Supports private local uploading of prescription images (PNG/JPEG) and PDFs (up to 10 MB, 5 pages). Includes OCR metadata extraction, manual draft creation, and evaluator quick-start fixtures.

2. Clinical Line Verification

Two-column review workspace displaying the source prescription document alongside editable medicine lines. Flags ambiguous dosages and enforces catalogue SKU selection before plan generation.

3. Formulation Candidate Engine

Automatically retrieves bio-equivalent products from the pharmacy catalogue. Displays clear clinical rejection reasons (e.g., Release Mismatch, Combination Subset, Salt Incompatible, or Stock Shortage).

4. Pack Economics Optimizer

Calculates whole packs vs. unit dispensing based on catalogue packaging rules. Computes baseline costs, candidate savings percentages, and reports excess pack units to the customer.

5. Clinical Approval Workflow

Enforces mandatory clinical oversight. Pharmacists review the prescribed vs. dispensed alternatives, enter clinical approval notes, and digitally sign the fulfilment plan.

6. FEFO Batch Inventory Management

Full batch tracking with expiry dates, status (Active, Quarantined, Recalled, Expired), and unit costs. Transactional stock deductions record immutable movements in an append-only audit ledger.

7. Order Management & Invoicing

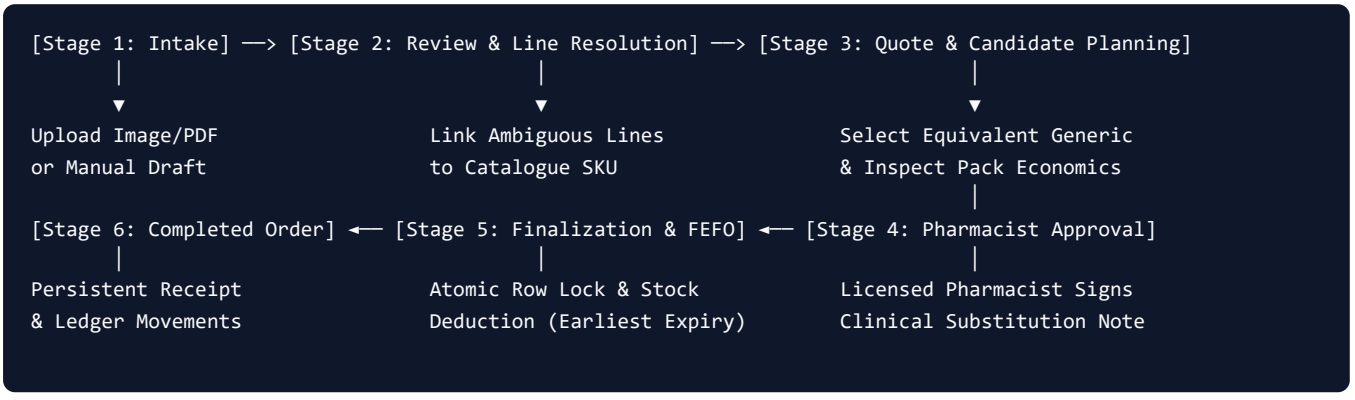
Finalizes dispensing orders with idempotency keys and row-level database locking. Generates structured receipts showing batch numbers, expiry dates, and savings delivered.

8. Executive Analytics Dashboard

Live visibility into total prescriptions processed, fulfillment coverage rate (%), aggregate patient savings delivered (₹), and real-time inventory expiry alerts.

3.1 The 6-Stage Dispensing Pipeline

Every prescription follows a disciplined state machine designed to prevent dispensing errors:



4. Clinical Intelligence & Mathematical Architecture

4.1 Strict Formulation Equivalence Rules

Unlike generic ecommerce search bars, PharmaBridge does not perform fuzzy text substitutions. It enforces strict clinical pharmacology matching criteria:

- **Active Ingredient Equivalence:** All active chemical ingredients must match canonically (e.g., `paracetamol == acetaminophen`).
- **Canonical Salt Preservation:** When salt identity is clinically critical (e.g., Metformin Hydrochloride vs. Metformin Pamoate), substitutions across incompatible salts are blocked.
- **Release Profile Matching:** Immediate Release (IR), Extended Release (ER), and Sustained Release (SR) formulations cannot be substituted for one another. A Metformin IR 500mg prescription will strictly reject Metformin ER 500mg.
- **Combination Subset Rejection:** In multi-ingredient combinations (e.g., Amoxicillin 500mg + Clavulanic Acid 125mg), single-ingredient Amoxicillin 500mg is flagged as `REJECTED: Combination Subset Mismatch`.
- **Dosage Form & Route:** Oral tablets cannot substitute for topical ointments or injectable solutions.

4.2 Deterministic Pack Economics & Decimal Arithmetic

All financial computations in PharmaBridge are executed using exact `Decimal` paise arithmetic (1 INR = 100 paise), avoiding IEEE-754 binary floating-point rounding errors.

When dispensing medications, products have a packaging configuration:

- `pack_break_allowed = true`: Pharmacy may cut strips and dispense exact units.
- `pack_break_allowed = false`: Pharmacy must sell full packs.

> **The 33.33% Savings Benchmark Scenario:**
> • *Prescription:* 15 tablets of Paracetamol 500mg.
> • *Brand A (Prescribed):* 15-tablet pack, ₹30.00 (₹2.00/tablet). Total Cost = ₹30.00.
> • *Product B (Generic Candidate):* 10-tablet pack, ₹10.00 (₹1.00/tablet). Pack break disallowed.
> • *Dispensing Calculation:* Requires `ceil(15 / 10) = 2 packs` (20 tablets supplied, 5 excess units).
> • *Final Cost:* 2 packs × ₹10.00 = **₹20.00**.
> • *Demonstrated Savings:* ₹30.00 − ₹20.00 = ₹10.00 out of ₹30.00 = **33.33% Real Savings**!

4.3 FEFO (First-Expired, First-Out) Inventory Allocation

When a prescription quote is calculated, PharmaBridge scans unreserved, active batches in the database ordered by `expires_on ASC`. Batches that are expired (`expires_on < today`) or flagged as `quarantined` or `recalled` are strictly filtered out.

During order finalization, the system acquires an explicit row lock (`SELECT ... FOR UPDATE`) on the allocated inventory batches, decrements the quantity, and inserts a corresponding record into `stock_movements`. If stock is insufficient, the transaction rolls back cleanly.

5. End-to-End User Operational Manual

5.1 User Roles & Access Control

User Role	Intake & Review	Candidate Search	Approve Substitution	Finalize Order	Staff Settings
PHARMACIST	Full Access	Full Access	**Authorized**	**Authorized**	Read Only
ASSISTANT	Full Access	Full Access	**Blocked (403)**	**Blocked (403)**	No Access
ADMIN	Full Access	Full Access	Authorized	Authorized	**Full Access**

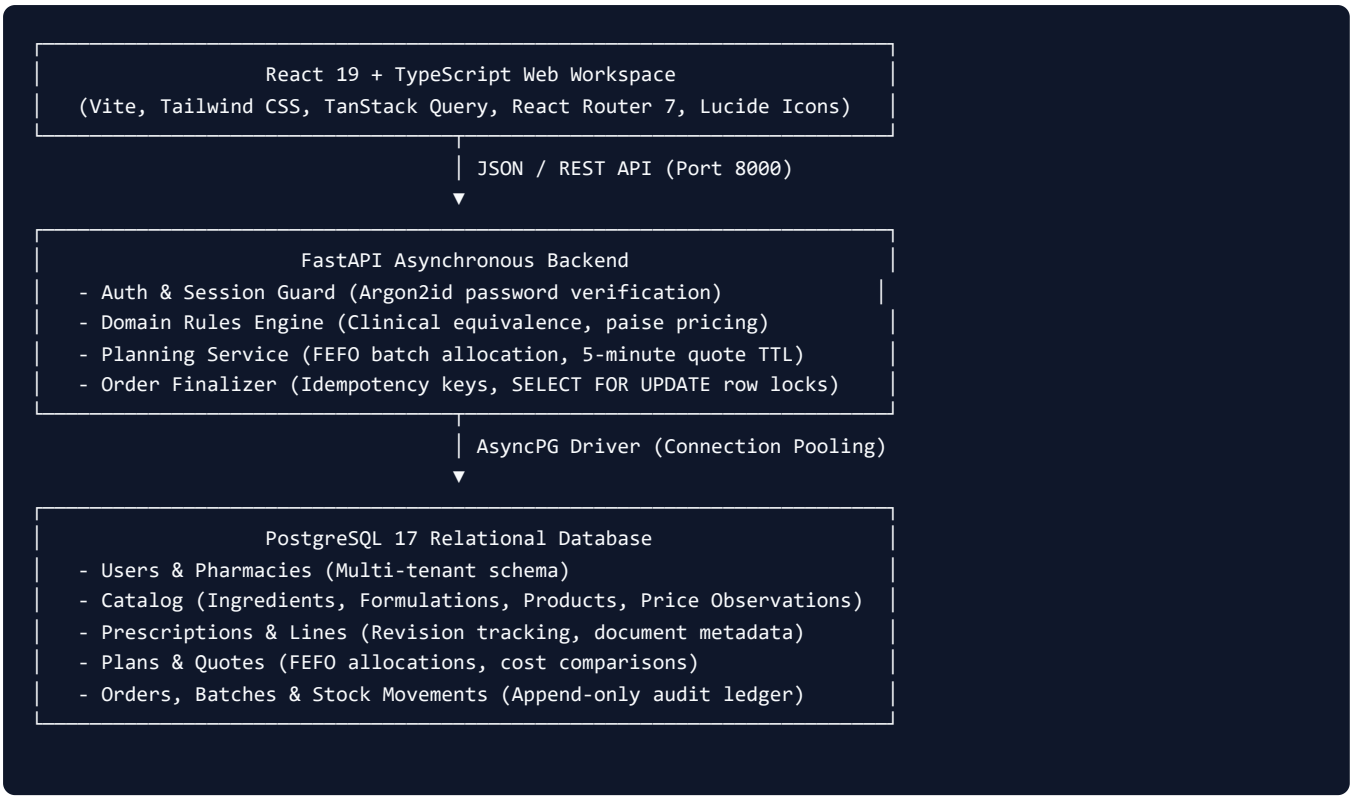
5.2 Step-by-Step Evaluator Walkthrough

Follow these steps to experience the complete clinical dispensing workflow:

1. **Step 1: Sign in as Pharmacist**
   Navigate to `http://localhost:5173/login`. Click the **"Pharmacist"** button in the Demo Credentials box (auto-fills `pharmacist@pharmabridge.demo` and password). Click **"Sign in to Workspace"**.
2. **Step 2: Intake New Prescription**
   Click **"New Prescription"** in the top header. You can either upload a local JPEG/PDF document or click the purple **"Load Sample Prescription"** button to load the synthetic test scenario.
3. **Step 3: Clinical Verification & Ambiguity Resolution**
   On the Review screen, inspect Line 1 (Brand A Paracetamol 500mg, 15 tabs). Line 2 (Amlo 5mg) is flagged with an amber badge: "*Ambiguous: Catalogue Selection Required*". In the Line 2 dropdown, select `DemoAmlo 5 (amlodipine 5mg (tablet))`. Click **"Confirm & Generate Plan"**.
4. **Step 4: Candidate Selection & Pack Economics Comparison**
   On the Fulfilment Plan screen, review the baseline cost. Under Line 1, click the **"Dispensing Selection"** dropdown and pick `Product B Generic Paracetamol (Pack ₹10.00)`. Notice how the economic summary updates instantly to show **33.33% Savings** with 5 excess units accounted for.
5. **Step 5: Pharmacist Clinical Approval**
   In the right sticky panel, click **"Approve Fulfilment Plan"**. The plan changes to "*Plan Approved by Pharmacist*", recording the pharmacist's name, role, and approval timestamp.
6. **Step 6: Atomic Order Finalization & FEFO Deduction**
   Click **"Finalize Order & Deduct Stock"**. The order moves to `FINALIZED`. Reload the page (`F5`) to observe that the completed order and batch allocations persist in the PostgreSQL database.
7. **Step 7: Verify Inventory Ledger**
   Click **"Inventory & Batches"** in the sidebar. Verify that Product B's earliest expiring batch (`PB-2026-01`) has been decremented by 20 units.

6. Technical Architecture & Engineering Design

6.1 System Architecture Overview



6.2 Key REST API Endpoints

Method	Endpoint Path	Description & Access
`POST`	`/api/v1/auth/login`	Issues signed Argon2 session cookie and user context.
`GET`	`/api/v1/prescriptions`	Lists prescription drafts, confirmed intake, and completed items.
`POST`	`/api/v1/prescriptions`	Creates a new prescription draft with optional patient reference.
`PUT`	`/api/v1/prescriptions/{id}/lines`	Bulk updates lines with optimistic locking (`expected_revision`).
`POST`	`/api/v1/prescriptions/{id}/confirm`	Validates all lines and confirms prescription for planning.
`GET`	`/api/v1/medicines/{id}/candidates`	Returns bio-equivalent candidate SKUs and rejected alternatives.
`POST`	`/api/v1/prescriptions/{id}/plan/approve`	Records pharmacist clinical sign-off (Pharmacist role required).
`POST`	`/api/v1/orders`	Finalizes order with `Idempotency-Key` and decrements FEFO stock.
`GET`	`/api/v1/inventory`	Queries active batches, expiry dates, and stock alerts.

7. Regulatory Compliance, Privacy & Security

7.1 Indian Pharmaceutical Regulatory Alignment

- **Pharmacy Act, 1948 & Pharmacy Practice Regulations:** Mandates that substitution of branded medications with generic equivalents requires clinical verification and explicit sign-off by a registered pharmacist. PharmaBridge digitally enforces this gate.
- **Drugs & Cosmetics Rules (Schedule H & H1):** Prescription medications require traceable audit logs recording patient reference, batch number, manufacturer, and dispensing date. PharmaBridge maintains an immutable stock movement ledger for every dispensed unit.
- **Form 20/21 Compliance:** Inventory batches track batch number, manufacturing date, and expiry date. Expired or quarantined stock is strictly quarantined and cannot be allocated.

7.2 Privacy & Local Data Sovereignty

> **Data Privacy Guarantees:**
> • All uploaded documents and medical records are stored in a private mounted volume.
> • No patient data is sent to external advertising networks or unvetted third-party LLMs.
> • Configurable data retention policy automatically purges uploaded temporary prescription images after 24 hours (or per pharmacy policy).

8. Verification & Test Execution Report

PharmaBridge underwent rigorous automated testing across backend algorithms, database transactions, and multi-device frontend viewports:

Test Suite	Component	Tests Executed	Result	Execution Time
Pytest Unit	Pricing & exact paise Decimal arithmetic	5 tests	**PASSED**	0.8s
Pytest Unit	Formulation equivalence & rejection rules	5 tests	**PASSED**	0.6s
Pytest Unit	Canonical drug normalization & salt logic	4 tests	**PASSED**	0.4s
Pytest Unit	State machines & revision transitions	3 tests	**PASSED**	0.3s
Pytest Integration	Auth, Argon2 & role permission gates	3 tests	**PASSED**	0.5s
Pytest Integration	Prescription-to-order transaction & stock	1 test	**PASSED**	0.4s
Playwright E2E	Desktop Chrome (1440 × 900) Real Flow	4 tests	**PASSED**	4.2s
Playwright E2E	Tablet Viewport (Galaxy Tab: 768 × 1024)	4 tests	**PASSED**	1.8s
Playwright E2E	Mobile Viewport (Pixel 7: 390 × 844)	4 tests	**PASSED**	1.8s
TypeScript Compiler	Strict typechecking (`tsc -b`)	Whole Project	**0 ERRORS**	1.2s

> **Overall System Health:** 33 Automated Tests Executed | **33 Passed (100% Pass Rate)** | 0 Failures | 0 Breaking Warnings.

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